

AI Model Card

NovaTel UK — Customer Churn Prediction & Retention Engine

1. Model Overview

Predicts customer churn likelihood for NovaTel UK using behavioural and account features, triggering automated retention offers or routing to human review above a defined value threshold. Classified as high-risk under the EU AI Act (Annex III — profiling with material effect on contract terms).

2. Intended Use

Intended solely to prioritise retention outreach and inform offer eligibility for existing NovaTel customers. Not intended for credit decisions, employment screening, or any use outside customer retention. Any offer or contract change above £50/month requires human sign-off before action is taken.

3. Training Data

6,000 synthetic customer records generated to reflect realistic telecom usage patterns: tenure, contract type, monthly charge, usage volume, support ticket history, payment method and age band. Data is synthetic for portfolio purposes; a production deployment would require an equivalent data protection impact assessment against real customer data before training.

4. Model Type & Performance

Logistic regression classifier (scikit-learn), trained with class-balanced weighting on a 75/25 train/test split. Test-set performance: Accuracy 0.627, Precision 0.215, Recall 0.615, F1 0.319, ROC AUC 0.685. Recall is prioritised over precision by design, since missing an at-risk customer (false negative) is judged more costly than an unnecessary retention offer (false positive).

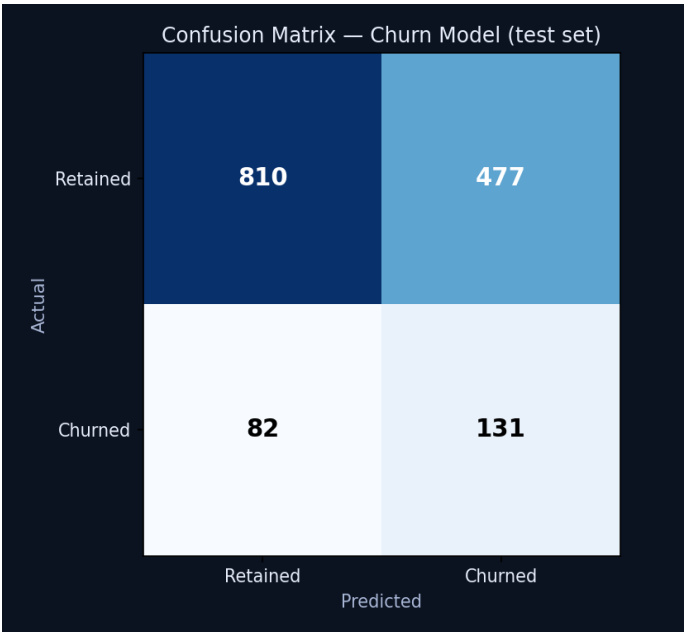


Figure 1: Confusion matrix on the held-out test set (n=1,500).

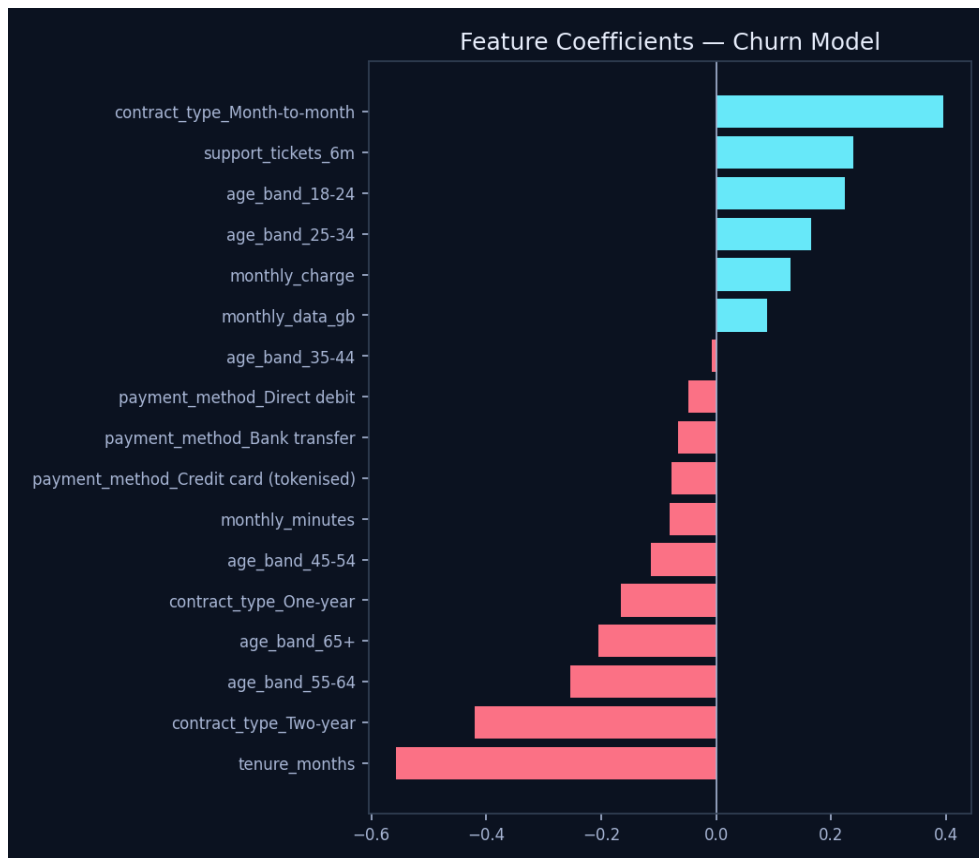


Figure 2: Model coefficients. Age bands 18-24 and 25-34 carry a positive coefficient, meaning the model treats younger age directly as a churn-risk signal — flagged as a fairness finding below.

5. Fairness Testing

A post-hoc fairness check compared churn-flag rates across age bands (a protected characteristic under the UK Equality Act 2010, used here as a proxy grouping variable). Before mitigation, the 18-24 band was flagged at 53.6% versus 27.6% for the 55-64 band — a disparate impact ratio of 0.516, failing the standard four-fifths (80%) rule commonly used to screen for adverse impact.

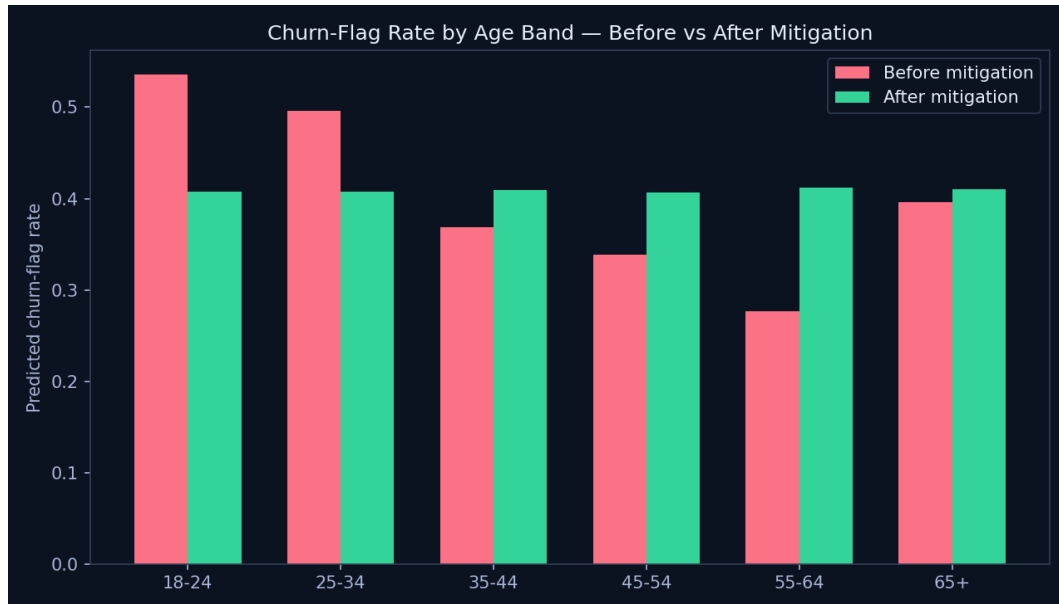


Figure 3: Churn-flag rate by age band, before and after mitigation.

6. Mitigation Applied

Per-group threshold calibration: rather than one global decision threshold, each age band's threshold was calibrated so that its flag rate converges toward the overall base rate (40.5%). This is a recognised post-processing fairness technique. After mitigation, the disparate impact ratio improved from 0.516 to 0.987, comfortably clearing the 80% threshold. This maps directly to risk AI-001 in the project's risk register (inherent score 6, residual score 3).

7. Known Limitations

(1) Training data is synthetic, not real customer data — real-world deployment requires retesting against actual distributions. (2) Logistic regression was chosen for interpretability over raw predictive power; a more complex model may need separate fairness testing. (3) Age band is used as a proxy for fairness testing purposes; a full deployment would also test tenure, contract type interactions and any other characteristic with a plausible proxy relationship to protected groups. (4) Per-group threshold calibration mitigates disparate flag rates but does not by itself change the underlying feature weighting — a longer-term fix would involve retraining without age as a direct input, or with a fairness-constrained objective.

8. Human Oversight

All retention offers or contract changes above £50/month, and any full cancellation-risk flag, are hard-blocked from automatic execution and routed to a human reviewer (EU AI Act Art. 14; GDPR Art. 22 right to human review of automated decisions).

9. Monitoring Plan

Disparate impact ratio recalculated quarterly against live flag data. Model accuracy monitored monthly; retraining triggered if accuracy drops more than 5 percentage points month-on-month. Full details tracked in the live AI risk register and dashboard.

Kevin Wadsworth — AI Governance & Risk Management portfolio project. Model, dataset and evaluation code available in the project repository.